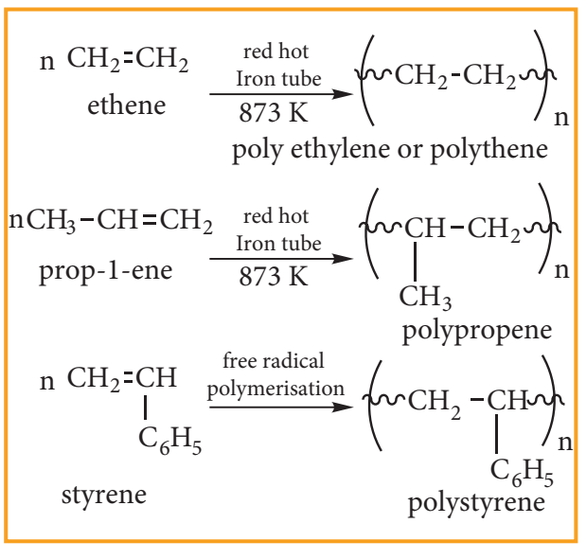
(iv) Polymerisation:

A polymer is a large molecule formed by the combination of larger number of small molecules. The process in known as polymerisation. Alkenes undergo polymerisation at high temperature and pressure, in the presence of a catalyst.

for example



Recycling plastics

Extensive use of polymers clogs up landfills and polute the environment. Because of diversity of polymers in consumer products, recycling requires sorting the polymers into various sub-types, labels with codes and symbols, which are then recycled separately.

Table shows the codes and symbols used in recycling of ethene-based additionpolymers.

(Lower the number, greater the ease of recycling the material)

CODE	TYPE	NAME	EXAMPLES
PET	1	Polyethylene terephthalate	Soft drinks bottles, jars, vegetable oil bottle.
HDPE	2	High-density polyethylene	Milk, water and juice containers
PVC	3	Polyvinyl chloride	Shampoo bottles, plastic pipes
LDPE	4	Low density polyethylene	Sandwich bags, grocery bags
PP	5	Polypropylene	Straws, diaper, toys
PS	6	Polystyrene	Disposable utensil, foam cups
Other	7	Multilayer plastics	Various flexible item.

13.3.4. Uses of Alkenes

1) Alkenes find many diverse applications in industry. They are used as starting materials in the synthesis of alcohols, plastics, liquors, detergents and fuels

2) Ethene is the most important organic feed stock in the polymer industry. E.g. PVC, Sarans and polyethylene. These polymer are used in the manufacture of floor tiles, shoe soles, synthetic fibres, raincoats, pipes etc.,

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